Efficient Fine-Tuning with BERT Adapters: A Network Science Perspective

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Abstract

As language models continue to scale, full parameter fine-tuning becomes increasingly computationally expensive. Bottleneck adapters offer a parameter-efficient alternative by inserting small, trainable modules within the layers of a pre-trained model. In this paper, we investigate the effectiveness of BERT adapters for topic classification in network science research. We characterize the performance-compute trade-off and demonstrate that adapters can achieve competitive results while training less than 1% of the original model parameters.

1. Introduction

Large-scale language models (LLMs) have revolutionized natural language processing (NLP). However, as mentioned in Devlin et al. (2018), full parameter fine-tuning requires significant computational resources, especially when dealing with domain-specific datasets in fields like Network Science.

2. Methodology

We utilize the bottleneck architecture described in Houlsby et al. (2019). This approach inserts two modules per transformer layer: one after the multi-head attention and one after the feed-forward layer. Each module consists of a down-projection, a non-linear activation, and an up-projection, effectively reducing the number of trainable parameters while maintaining representation quality.

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3. Results and Discussion

Our experiments on the CSDS research dataset indicate that BERT adapters converge faster than full fine-tuning. We find that the structural properties of the attention weights in adapted models exhibit higher modularity compared to their fully fine-tuned counterparts, suggesting that adapters preserve the original model's "backbone" more effectively.

4. Conclusion

We have demonstrated that BERT adapters are a highly efficient tool for scientific development in NLP. By minimizing the trainable parameter count, we enable more rapid iteration and experimentation on academic-scale hardware.

5. References

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